

Prestige and behavior distortion: the effect of MICHELIN awards on restaurant ratings and reviews

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Abstract

We study how a sharply-timed prestige shock, the receipt of a MICHELIN award, affects reviewer behavior on the online review platform Google Reviews. Using an incomplete panel of MICHELIN starred restaurants in New York, Chicago, and California matched to local Google review data, we exploit the precise timing and exogeneity of the award announcement in a regression discontinuity design with time as the running variable. We find negative but statistically insignificant effects on ratings and positive, significant effects on review volume in the months following announcement. The rating response is driven by “heavy” reviewers—those with five or more MICHELIN reviews, whose ratings turn more negative immediately after announcement, which we argue is consistent with an unobservable difference in expectations channel in which prestige raises the benchmark against which the experience is judged. A difference-in-discontinuities design intended to net out a common ceremony effect proves inconclusive, and we argue its robustness trades off precision in a way which favors our baseline discontinuity design.

1 Introduction

Awarded annually by the MICHELIN Guide for culinary excellence, MICHELIN stars are considered the most prestigious award in the restaurant industry. While the MICHELIN guide does not keep cumulative historical records, one report estimates there are around 3,000 establishments across the world with at least one MICHELIN star (Swiss Education Group, 2026). To a restaurant aiming for MICHELIN recognition, a star is a pure, sharply-timed prestige shock which immediately changes the way patrons perceive and therefore experience the establishment despite no immediate and accompanying change in menu prices or fare quality.

Understanding how this change in perceived prestige affects diner behavior provides insight into how intangible characteristics of patrons’ dining experiences (i.e., the opinion of others) manifest in their own experiences and thus, indirectly, demand for that same experience by others. Analysis of the effect of Yelp reviews on restaurant outcomes from Luca (2016) finds a one star Yelp rating increase to cause a 5-9% increase in revenue. The fact Luca finds diner attitudes to be sensitive to authoritative reviewer identity (e.g., “elite” Yelp reviewers) motivates our hypothesis of diner sensitivity to MICHELIN recognition. Conversely, a naive comparison analysis from Enigma (2025) found only substantial gains from for three star establishments (median \$1.9M annual revenue) and moderate gains for

*Coding was done in collaboration with Claude Code and OpenAI Codex. All views and errors are my own.

two star establishments (median \$1.4M annual revenue). Gains for one star, Bib Gourmand, and “selected” (restaurants recognized by the MICHELIN guide but not deemed worth of stars) were negligible (\$1.4M vs \$1.2M vs \$1.1M, respectively), with differences possibly driven by differences in patronage. The broader context of our question is thus the following: downstream use of the prestige signal depends strongly on whether the signal itself is present, which is what we aim to detect.

To answer this question, we leverage the precise timing and exogeneity of the award announcement in a regression discontinuity design (RDD) to identify the local effect of MICHELIN award receipt on restaurant review volume and ratings and find negative but insignificant effects on ratings and positive and significant effects on review volume. The pooled effect on ratings appears to be driven by heavy reviewers, reviewers with five or more MICHELIN reviews in our observed time window, whose ratings shortly after award announcement are much more negative (and significant) relative to their casual peers. We use the difference-in-discontinuity (DIDC) design of Grembi, Nannicini, and Troiano (2016) to difference out shocks with common timing as the MICHELIN award announcement, comparing restaurants with a change in MICHELIN award status to those without. Evidence from this exercise is relatively inconclusive, though we argue the robustness-precision trade off demanded by DIDC supports our original RDD empirical strategy.

The rest of the paper proceeds as follows. Section 2 walks through the MICHELIN award process from the perspective of a fictional restaurant, *De Luxe*. Section 4.1 develops a simple structural model to motivate the empirical strategy presented in section 4.2. Section 5.1 presents headline results, and we explore possibility heterogeneity across reviewers in section 5.2. Finally, section 6 concludes.

2 Background

For simplicity, we discuss the process a restaurant undergoes to obtain one MICHELIN star.¹ The process is largely similar for a restaurant’s receipt of 2 and 3 stars. We use a fictional fine dining restaurant, *De Luxe*, to illustrate the fact the MICHELIN star award process is highly exogenous.

MICHELIN Guide (2025b) evaluates restaurants in select geographies according to five published criteria: ingredient quality, mastery of cooking techniques, flavor harmony, the personality of the chef expressed through the cuisine, and consistency across visits and menu items. *De Luxe* understands these criteria and incorporates them into operational decisions (e.g., staffing, layout, menu pricing). The degree to which these criteria are satisfied determine not only whether or not *De Luxe* will receive a MICHELIN award but how many (restaurants can earn more than 1 star) they will receive.

Shortly after opening to the public, *De Luxe* engages a hospitality PR firm to generate a critical mass of food media coverage getting them on the radar of MICHELIN inspectors

¹The process is slightly different for receipt of a MICHELIN Bib Gourmand award, but we focus on MICHELIN star(s) due to the fact most of the restaurants in our sample change star status (0 to 1 star, 2 to 3 star, etc.) and not Bib Gourmand status.

who, like every other diner, choose where to eat. Importantly, *De Luxe* nor their PR firm pitches to MICHELIN as the MICHELIN guide does not accept pitches.

The first milestone for *De Luxe* is press from a major critic platform like the New York Times, which is considered one of the highest visibility events in a young restaurant's life. *De Luxe* is aware of the New York Times critic's presence because the critic's article is not written or published anonymously.² However, even after reaching this milestone, *De Luxe* may have to wait months or years before being visited by MICHELIN inspectors as they are highly heterogeneous across time and geography. For example: New York's Atomix, opened in 2018, was awarded two stars three years after opening; New York's Saga, opened in 2021, earned two stars within one year; secondary reports suggest Dallas's Mamani, opened in 2025, clinched a star within two months.

Sometime in their second year open, *De Luxe* begins to be visited by inspectors. Critically, MICHELIN inspectors are anonymous, pay full price, and behave as ordinary guests, so *De Luxe* cannot control the table the inspector sits at, which dishes are ordered, nor whether the visit lands on a quiet Tuesday or a crushed Saturday (MICHELIN Guide, 2025a). Over the course of the next year, multiple inspectors from different regions will visit *De Luxe* multiple times.³

In the fall, the chef notices a party of four that lingered unusually long over a cheese course and asked specific, technical questions about a single supplier. The bill was paid without additional comment, and the chef's suspicion goes unconfirmed. This is the extent to which *De Luxe*, like many of its peer establishments pursuing MICHELIN recognition, can "know" and observe inspectors.

The award cycle is on an annual basis. After inspectors have enough information on *De Luxe*, which often takes a full year to collect, they deliberate internally. Shortly before the award ceremony, *De Luxe* is contacted privately by the MICHELIN Guide and are told only that they will be receiving MICHELIN star recognition. They are not told how many stars they will be receiving. Demotions (e.g., from 2 to 1 star) are not pre-notified privately in a similar way. Restaurants typically learn of their loss at the public announcement.

3 Data

We construct an incomplete panel from 2005 through 2026 of MICHELIN starred restaurants across the three oldest MICHELIN guide regions in the United States: New York, Chicago, and California.⁴ The panel contains data at the restaurant-year level, where the year corresponds to the date at which the awards were announced and not the date at which the guides were published. Local Google Review data comes from an extract com-

²Pete Wells was the New York Times's most recognizable food critic, who retired in 2024. Given his reputation, he is sure to be recognized by any staff member of any restaurant he visited.

³The MICHELIN Guide does not publish a fixed minimum number of visits, inspectors, or visits per inspector. One can plausibly assume these quantities differ across restaurants and geographies.

⁴California is the current guide for two subguides in San Francisco (started in 2006) and Los Angeles (started in 2008) which were merged together in 2019 when MICHELIN expanded a guide to the entire state of California. For the purpose of our analysis, we focus on establishments in San Francisco and Los Angeles.

piled by Li, Shang, and McAuley (2022) and Yan, He, Li, Zhang, and McAuley (2023) (which we refer to onward as the Li dataset), which contains all reviews for restaurants in select cities through 2021 at the rating-reviewer-timestamp level.

To construct the analysis-ready incomplete panel, we match 73.6% (379) of restaurants in the MICHELIN panel to the Li dataset. We are unable to match the remaining 26.4% (136) due the restaurant in the MICHELIN panel being absent in the local review data due to closure or not yet being open. Importantly, our analysis may be biased by survivorship due to missingness of awardee restaurants due to closure.

We define an event as the day the MICHELIN guide announced a restaurant changing award status, including both promotions (e.g., from 0 to 1 star) and demotions (e.g., from 3 to 2 stars). Most events are promotions from zero to 1 star (93, 73.8%), followed far behind by direct promotions from zero to 2 stars (13, 10.3%) or 1 to 2 stars (8, 6.3%). Demotions account for a small portion of events (pooled 6 events, 4.8%).

Table 1 presents descriptive statistics on reviewer behavior before and after announcement of the MICHELIN award. Our effective sample spans 126 MICHELIN award-change events for 116 restaurants across 2015 through early 2020, a time window containing the bulk of the reviews in the Li dataset. At first glance, restaurants receive slightly more reviews ($\Delta = 0.266$, significant at $q = 0.1$ Benjamini-Hochberg adjusted 0.05-level) and negligibly more negative ratings ($\Delta = -0.008$, indistinguishable from 0 at $q = 0.1$ Benjamini-Hochberg adjusted 0.05-level) after receiving a MICHELIN award.

Table 1: Pre/post-treatment outcomes for treated restaurants (status-change events)

	<i>Pre-treatment</i>				<i>Post-treatment</i>				<i>Difference in Means</i>	
	Mean	Q1	Median	Q3	Mean	Q1	Median	Q3	Δ	SE
Review count	1.466	0.394	0.923	1.721	1.731	0.593	1.148	2.037	0.266	0.076
Mean rating	4.571	4.443	4.613	4.750	4.564	4.465	4.614	4.752	-0.008	0.032

Notes. Sample restricted to restaurants experiencing status-change events. Pre-treatment is defined as weeks $[-26, -1]$; Post-treatment is defined as weeks $[0, 26]$, with week 0 counted as Post. The pre-treatment N counts all events within the ± 26 -week window; the post-treatment N restricts to events passing the post-active filter. Quartiles summarize event-level pre- and post-treatment outcomes. Sample 2016–2019. The mean rating row uses the $n = 108$ events with rateable observations in both windows. Bold Δ denotes $p < 0.05$ under a paired t test and survives Benjamini–Hochberg FDR control at $q = 0.10$. Pre-treatment award composition: 1★ 77.8% ($n = 98$), 2★ 19.0% ($n = 24$), 3★ 3.2% ($n = 4$).

4 Model

4.1 Conceptual framework

Consider a market of restaurants indexed by $j = 1, \dots, J$ visited by diners indexed $i = 1, \dots, n_j$ across time t . If T_j is the date at which restaurant j receives MICHELIN recognition, we define the treatment indicator $M_{jt} = \mathbf{1}\{t \geq T_j\}$ so MICHELIN recognition turns

on discontinuously at the date of announcement.⁵

Restaurants have latent quality q_{jt} , price point p_{jt} , and perceived prestige ψ_{jt} . We model perceived prestige as a combination of review volume, average rating, and MICHELIN award status:

$$\psi_{jt} = \rho \log N_{j,t-1} + \eta R_{j,t-1} + \delta M_{jt}, \quad (1)$$

where $N_{j,t-1}$ and $R_{j,t-1}$ are the lagged review volume and average rating of restaurant j . The parameter $\delta > 0$ is the shock to a restaurant's prestige upon receiving a MICHELIN award. Under the assumption accumulated attention from review volume and average ratings boost the perceived prestige of restaurant j , we assume $\rho, \eta \geq 0$. Importantly, we distinguish between the restaurant-level, public signal of prestige ψ_{jt} and the diner-level response to that signal.

Diner i 's utility from visiting restaurant j at time t is

$$U_{ijt} = u(q_j) - p_{jt} + \alpha_i \psi_{jt} + \varepsilon_{ijt}, \quad (2)$$

where $u(q_j) - p_{jt}$ is the direct consumption value of restaurant quality scaled by its price point. α_i captures diner i 's valuation of prestige and ε_{ijt} is an idiosyncratic taste shock parameter with mean 0.

A diner chooses to restaurant j is $U_{ijt} \geq \bar{U}_i$, where $\bar{U}_i$ is the diner's reservation utility capturing alternative dining options or eating at home. Aggregate demand for restaurant j is $D_{jt} = D(p_{jt}, q_{jt}, \psi_{jt})$ with $\frac{\partial D}{\partial \psi} > 0$ so higher perceived prestige increases demand, holding price and quantity fixed.

Not all diners will leave a review of the restaurant. Suppose a share of diners $\pi(\psi_{jt}) \in (0, 1]$ leave reviews so the total number of reviews for restaurant j at time t can be expressed $N_{jt} = \pi(\psi_{jt}) D_{jt}$. We assume review propensity is increasing in perceived prestige (i.e., $\pi'(\psi_{jt}) \geq 0$), allowing receipt of a MICHELIN award to increase reviews both because more diners visit and because more diners are more likely to post about a prestigious restaurant.

Conditional on having visited and reviewed a restaurant, reviewer i 's true evaluation of restaurant j at time t is

$$R_{ijt}^* = q_{jt} + \theta_i \psi_{jt} + \xi_{ijt}. \quad (3)$$

The observed review is a monotone transformation of the true evaluation, $R_{ijt} = h(R_{ijt}^*)$, where R_{ijt} may be a five-level star rating, text sentiment, or another measure of review positivity. The parameter θ_i captures the effect of prestige on the reviewer's evaluation. Structurally, we can decompose additively so $\theta_i = \theta_i^{\text{taste}} + \theta_i^{\text{conform}}$. The first component θ_i^{taste} captures the reviewer's innate taste for prestige, such as a preference for fine dining or luxury experiences; the second component $\theta_i^{\text{conform}}$ captures the reviewer's preference to agree with the consensus, perhaps a social or psychological cost of disagreeing with the crowd (e.g., deviation from the average rating R_{jt}) or a prestigious expert signal (e.g.,

⁵Of course, we can allow for ordinal MICHELIN award status $M_{jt} \in \{1, 2, 3, 4\}$, where $M_{jt} = 1$ corresponds to a Bib Gourmand award and $M_{jt} \in \{2, 3, 4\}$ corresponding to $M_{jt} - 1$ MICHELIN stars.

deviation from MICHELIN award status M_{jt}). A simple reduced-form is the following:

$$R_{ijt} = q_{jt} + \theta_i \psi_{jt} + \xi_{ijt}. \quad (4)$$

If $\mathbb{E}[\theta_i \mid i \text{ reviews restaurant } j] > 0$, then MICHELIN recognition increases observed review positivity through the prestige channel. The main prediction of our structural model is the following:

Proposition 1 (MICHELIN recognition generates discontinuous review responses). *Suppose ex-MICHELIN prestige $\tilde{\psi}_{jt} = \rho \log N_{j,t-1} + \eta R_{j,t-1}$, restaurant quality q_{jt} , and price p_{jt} are continuous at time $t = T_j$. Assume review volume satisfies*

$$N_{jt} = g(\psi_{jt}, q_{jt}, p_{jt}) + \nu_{jt} \text{ with } \frac{\partial g}{\partial \psi} > 0,$$

with $\mathbb{E}[\nu_{jt}]$ continuous at T_j . Assume observed ratings satisfy

$$R_{ijt} = q_{jt} + \theta_i \psi_{jt} + \xi_{ijt} \text{ with } \mathbb{E}[\theta_i \mid i \text{ reviews restaurant } j] > 0,$$

with $\mathbb{E}[\xi_{ijt} \mid i \text{ reviews restaurant } j]$ continuous at T_j . Then MICHELIN recognition generates positive discontinuities in expected review volume and expected ratings at the award date. Formally,

$$\begin{aligned} \lim_{t \searrow T_j} \mathbb{E}[N_{jt}] - \lim_{t \nearrow T_j} \mathbb{E}[N_{jt}] &> 0 \\ \lim_{t \searrow T_j} \mathbb{E}[R_{ijt}] - \lim_{t \nearrow T_j} \mathbb{E}[R_{ijt}] &> 0. \end{aligned}$$

Proof. See appendix A. $\square$

4.2 Empirical strategy

A natural application of the the structural model outlining the conceptual framework in section 4.1 is a sharp RDD with time as the running variable. Let event time for restaurant j be $\tau_{jt} = t - T_j$. The baseline estimating equation is the following:⁶

$$Y_{jt} = \alpha_j + \delta_1 \mathbf{1}\{\tau_{jt} \geq 0\} + \delta_2 \tau_{jt} + \beta \mathbf{1}\{\tau_{jt} \geq 0\} \tau_{jt} + \varepsilon_{jt}, \quad (5)$$

where Y_{jt} is the relevant outcome variable $Y_{jt} \in \{R_{jt}, N_{jt}\}$. The specification above tests whether proposition 1 manifests empirically in our sample of MICHELIN restaurants.⁷ A positive estimate $\hat{\beta}$ on the interaction term is evidence supporting the structural prediction that MICHELIN awards cause discontinuous jumps in both review volume and ratings shortly after their announcement.

One can also apply a regression adjustment with restaurant-level covariates $\mathbf{X}_{jt}$ to

⁶Equation 5 yields estimates which are (under some additional conditions) equivalent to the non-parametric RDD estimates with a rectangular kernel, which is what we implement in R using Calonico and coauthors' (2015) package `rdrobust`. We explicate the linear model itself for clarity.

⁷However, due to limited data on relevant covariates, we do not apply such a regression adjustment.

reduce the noise in the causal parameter of interest $\hat{\beta}$. However, implementing such an adjustment is not possible to limited data on relevant covariates; instead, we residualize the outcome variable Y_{jt} on restaurant and award year fixed effects estimated using data from before the restaurant's award ceremony. Our baseline specification in equation 5 is then run with the residualized outcome variable $\tilde{Y}_{jt}$ in place of Y_{jt} :

$$\tilde{Y}_{jt} = Y_{jt} - \left[\bar{Y}_{j\bullet} + \sum_{t=1}^{T_j} \bar{Y}_{\bullet t} \right]$$

In the continuity-based RDD setting, identification leans on the assumption the relevant potential outcome in absence of the MICHELIN award $Y_{jt}(0)$ is continuous in time (Cattaneo, Idrobo, & Titiunik, 2019).

$$\lim_{t \searrow T_j} \mathbb{E}[Y_{jt}(0)|t] = \lim_{t \nearrow T_j} \mathbb{E}[Y_{jt}(0)|t].$$

Simply put, we assume reviewer behavior (e.g., review volume, average rating, etc.) evolves smoothly through the award announcement date in the absence of MICHELIN recognition. We argue this assumption is plausible in a narrow window around T_j as there are no other discontinuous shocks occurring exactly at or within a narrow window of the MICHELIN announcement date.

The review volume spike predicted by proposition 1 immediately following treatment will likely present as bunching, which is often considered to be evidence against the RD's identifying smoothness assumption. Individuals who dined at restaurant j prior to the MICHELIN award announcement who would not have posted a review absent the award may be incentivized to post a review in response to the prestigious award despite having not visited after the award announcement.

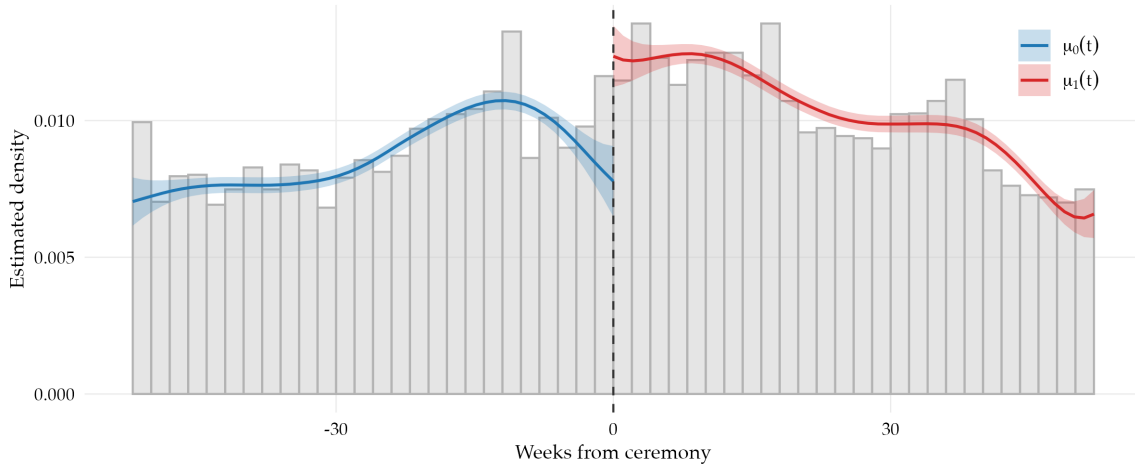


Figure 1: Cattaneo et al. (2018) density test around the MICHELIN award announcement date

Indeed, the density test from Cattaneo et al. (2018) fails at all conventional levels of

significance ($p = 2.3 \times 10^{-7}$), which gives merit to the bunching narrative we just posed. However, rather than view this as a genuine threat to identification, we embrace the test’s rejection of the null of no sorting as an inherently interesting result we discuss further in section 5.

The multimodal distribution presented in figure 1 suggests the jump in review volume or average rating could be a product of common time shocks to review volume itself shared by treated and no change MICHELIN establishments. One possibility, for instance, is the award ceremony and subsequent press driving greater attention broadly toward MICHELIN award establishments rather than a differential prestige shock confined only to newly awarded (or demoted) restaurants.

To test this concern, we implement the difference-in-discontinuities design first advanced by Grembi et al. (2016) to compare restaurants with a change in MICHELIN award status (treatment group) against those without a change (placebo group). DIDC, similar to difference-in-differences, will difference out any common “ceremony” effect which also reaches restaurants with no change in MICHELIN award status. Per Picchetti, Pinto, and Shinoki (2024), we obtain the DIDC estimate by differencing the discontinuities from equation 5 for both award change and no change restaurants.⁸

5 Results

Regression discontinuity estimates for responses to the MICHELIN award announcement on ratings and review volume are presented in tables 2 and 3, respectively. Columns (A) and (B) present bandwidths we contend correspond to windows in which the only plausible shock comes from the MICHELIN award announcement. Within a two week half-width window (a window of one month centered at the ceremony), an awardee can neither anticipate nor change in response to receipt of the award, supporting the plausibility of our design. However, descriptives in table 1 hint two weeks may be insufficient time to detect any changes in reviewer behavior. We argue similarly against anticipation and restaurant development for a one month half-width window (a window of two months centered at the ceremony), which may provide enough time for identifying variation to manifest in our sample. Columns (C) through (E) present estimates corresponding to MSE optimal bandwidth and multiples of it for robustness.

5.1 Headline Results

Point estimates for the discontinuity with ratings as the outcome variable are initially negative (-0.290 at $h = 14$ and -0.157 at $h = 30$, both indistinguishable from zero at conventional levels of significance) attenuate toward zero as bandwidth increases. Each of the 95% bias-adjusted confidence intervals comfortably straddle zero, with the estimate for the two week width being narrowly insignificant beyond the 0.05-level.

⁸The equivalence between the difference in the discontinuities and the discontinuity of the difference follows from claim 1 in Picchetti et al. (2024), with proof in the paper’s appendix C.

Table 2: Effect of Michelin Star-Status Changes on Star Rating — Review-Level Panel, All Reviewers

	(A) ± 14 days	(B) ± 30 days	(C) $\frac{1}{2} h^*$	(D) h^*	(E) $2 h^*$
Estimate	-0.290 (0.157)	-0.157 (0.123)	0.068 (0.074)	0.003 (0.048)	0.003 (0.044)
95% CI	[-0.597, 0.017]	[-0.399, 0.085]	[-0.078, 0.214]	[-0.092, 0.098]	[-0.084, 0.090]
p-value	0.064	0.203	0.359	0.953	0.945
N observations	817	1,749	5,248	10,147	18,029
N unique reviewers	842	1,735	4,919	9,322	15,825
Bandwidth	14.0 days	30.0 days	87.0 days	174.1 days	348.2 days

Notes: Estimates computed using `rdrobust` with triangular kernel and local polynomial order $p = 1$. Standard errors in parentheses are bias-adjusted clustered at the restaurant level. p -value tests the sharp null of no discontinuity at the cutoff against the two-sided alternative. h^* is the MSE optimal bandwidth. An observation is one individual review whose posting date falls within the bandwidth, the outcome is the per-review star rating (1-5), and the outcome is per-review star rating.

The negative estimates for the narrow bandwidths of two weeks and one month, together with the narrow insignificance of the estimate at $h = 14$, are unintuitive. On one hand, the negative estimate may be indicative of ex-MICHELIN shocks to rating behavior timed closely to the ceremony; on the other hand, the fact most restaurants receive 1-2 ratings per week also point to a narrative in which select restaurants coincidentally received particularly negative reviews around the time of their award receipt, biasing the estimate in the opposite direction. The first concern is precisely the worry motivating our DIDC specification.

The explanation we prefer is that the estimable change in ratings amounts to a difference between diners' expectation of the experience and the experience itself. To both casual and experienced reviewers alike, knowing a restaurant holds a MICHELIN star may raise expectations to a point that the difference between actual and expected experience becomes negative. Reversal of the effect over larger bandwidths follows from the possibility only reviewers dining at the restaurant immediately following the award announcement—perhaps those most eager to diner at, and therefore most critical of, the new awardee—exhibit this “elevated expectation” bias. Indeed, our hypothesis that restaurants do not change substantially within a local window of the award date may harm restaurants in the eyes of these types of reviewers. We comment further on this in our discussion of heavy reviewers in section 5.2.

The arguably more interesting pattern is the coefficient attenuation over larger bandwidths, which we take as evidence average ratings for newly minted MICHELIN restaurants increase gradually over time. Estimates for the MSE optimal bandwidth of around six months in column (D) and one year in column (E) are particularly suggestive of this smooth evolution in ratings, if at all present, as the point estimates are themselves close to zero and statistically indistinguishable from zero (0.003 with $p \approx 0.95$ for both) and

Contrary to our results on ratings, our findings on review volume construct a narrative aligned with our prediction from proposition 1. Expectedly, given the scale of the outcome

Table 3: Effect of Michelin Star-Status Changes on Review Volume — Daily Panel, All Reviewers

	(A) ± 14 days	(B) ± 30 days	(C) $\frac{1}{2} h^*$	(D) h^*	(E) $2 h^*$
Estimate	-0.004 (0.031)	0.091 (0.024)	0.062 (0.017)	0.073 (0.013)	0.060 (0.013)
95% CI	[-0.061, 0.061]	[0.010, 0.103]	[0.007, 0.073]	[0.017, 0.067]	[0.011, 0.062]
p-value	0.989	0.018	0.017	0.001	0.005
N observations	3,402	7,434	25,326	50,526	100,926
N unique reviewers	842	1,735	5,655	10,378	17,376
Bandwidth	14.0 days	30.0 days	93.7 days	187.5 days	374.9 days

Notes: Estimates computed using `rdrobust` with triangular kernel and local polynomial order $p = 1$. Standard errors in parentheses are bias-adjusted clustered at the restaurant level. p -value tests the sharp null of no discontinuity at the cutoff against the two-sided alternative. h^* is the MSE optimal bandwidth. An observation is a (restaurant $\times$ event $\times$ day) cell in the balanced daily panel within the running variable bandwidth, and the outcome is number of reviews.

variable, there is no detectable discontinuity (-0.004 with $p = 0.95$) within two weeks of award announcement. Within a month, however, the award announcement corresponds to an increase in 0.091 reviews. With an average of 0.18 to 0.22 (from one year to 90 days prior, respectively, both equivalent to around 1 review per week) reviews per day prior to ceremony, a newly minted MICHELIN restaurant can expect to see a jump of 1 additional review per week shortly after the ceremony. As bandwidth selection increases, the discontinuity estimates attenuate lightly but become increasingly distinguishable from zero, with the estimate of 0.073 at the MSE optimal bandwidth being significant at the 0.001-level.

But the question of whether the influx of reviews is caused by receipt of a MICHELIN award or by a broader “ceremony” effect remains unanswered. Table 4 presents discontinuity estimates from the specification in equation 5 being run on the sample of awarded (change in status) and non-awarded (no change in status) restaurants. Estimates for the no change subsample vary in sign across selected bandwidths but are all indistinguishable from zero. The difference between the awarded and no change subsample concur with our original discontinuity estimates from table 2 that there is no detectable sharp increase in ratings for awarded restaurants in response to a MICHELIN award. Notably, because both estimates are not significant at any conventional level of significance, we also cannot reject the hypothesis of no ceremony effect for the ratings outcome variable.

DIDC estimates on review volume in table 5 are more informative on the source of the prestige shock. Discontinuity estimates on the no change sample are—with one exception from the estimate run on the two week bandwidth—smaller in magnitude but share the same sign as the estimates on the award sample, though none are statistically significant. For example, awarded restaurants see a 0.091 review increase relative to the no change restaurants’ 0.005 increase following the announcement, producing a positive point estimate which is no longer distinguishable from zero at most conventional significance levels.

Together, the local effect of the Michelin announcement of 0.086 additional reviews per day under the one month bandwidth is positive but not distinguishable from zero. While these null results appear to support a ceremony effect rather than a Michelin effect, we

Table 4: Effect of Michelin Star-Status Changes on Star Rating — Review-Level Panel, All Reviewers (Difference-in-Discontinuities)

	(A) ± 14 days	(B) ± 30 days	(C) $\frac{1}{2} h^*$	(D) h^*	(E) $2 h^*$
<i>Awarded (β^A)</i>					
Estimate	-0.290 (0.157)	-0.157 (0.123)	0.068 (0.074)	0.003 (0.048)	0.003 (0.044)
95% CI	[-0.597, 0.017]	[-0.399, 0.085]	[-0.078, 0.214]	[-0.092, 0.098]	[-0.084, 0.090]
p-value	0.064	0.203	0.359	0.953	0.945
<i>No change (β^C)</i>					
Estimate	0.014 (0.094)	0.054 (0.060)	-0.052 (0.047)	-0.043 (0.042)	-0.021 (0.028)
95% CI	[-0.170, 0.199]	[-0.063, 0.171]	[-0.145, 0.040]	[-0.125, 0.039]	[-0.076, 0.035]
p-value	0.880	0.366	0.268	0.308	0.465
<i>Difference (β^{DIDC})</i>					
Estimate	-0.305 (0.183)	-0.211 (0.137)	0.121 (0.088)	0.046 (0.064)	0.024 (0.053)
95% CI	[-0.663, 0.054]	[-0.480, 0.057]	[-0.052, 0.293]	[-0.080, 0.171]	[-0.079, 0.127]
p-value	0.096	0.123	0.172	0.476	0.652
N observations (awarded)	817	1,749	5,248	10,147	18,029
N observations (no change)	4,335	9,680	29,003	56,992	106,285
N unique reviewers	4,910	10,364	28,696	53,653	65,180
Bandwidth	14.0 days	30.0 days	87.0 days	174.1 days	348.2 days

Notes: Estimates are computed by differencing two local linear regression discontinuity estimators $\beta^A - \beta^C = \beta^{DIDC}$, where β^A comes from awarded restaurants and β^C comes from same region, no change restaurants (no status change but same ceremony date). Both single sample fits use `rdrobust` triangular kernel with local polynomial order $p = 1$. The two fits share the same bandwidth, where h^* is the MSE optimal bandwidth from the awarded sample. Inference on DIDC parameter β^{DIDC} follows Picchetti et al. (2024).

instead interpret the imprecision of the differenced estimate as evidence of a failed attempt to protect against a temporal confounder (the award ceremony) which does not appear to have a significant effect on review volume on no change establishments. Critically, we advance this position into our heterogeneity analysis in the section that follows.

5.2 Heterogeneity

The conceptual framework from section 4.1 allows the prestige parameter θ_i to differ by reviewer, and one can plausibly assume prestige preferences to differ between diners who occasionally versus frequently visit MICHELIN establishments (which we proxy by MICHELIN review volume mapped to individual reviewers). We refer to this second group of individuals as “heavy” MICHELIN reviewers, defined as diners with five or more reviews across 2015 through 2019 written and posted on a MICHELIN restaurant.

Results in table 6 present a narrative which corroborates that told by the pooled estimates in table 2. Compared our headline point estimates, which were negative in small bandwidths and positive in larger ones (all indistinguishable from zero), discontinuity estimates are generally more negative across all bandwidths (with one anomalous positive discontinuity at half MSE optimal bandwidth) with varying levels of significance. In particular, estimates for the tighter bandwidths of two weeks and one month are better

Table 5: Effect of Michelin Star-Status Changes on Review Volume — Daily Panel, All Reviewers (Difference-in-Discontinuities)

	(A) ± 14 days	(B) ± 30 days	(C) $\frac{1}{2} h^*$	(D) h^*	(E) $2 h^*$
<i>Awarded (β^A)</i>					
Estimate	-0.004 (0.057)	0.091 (0.043)	0.062 (0.031)	0.073 (0.023)	0.060 (0.023)
95% CI	[-0.115, 0.108]	[0.007, 0.174]	[0.002, 0.122]	[0.028, 0.118]	[0.015, 0.105]
p-value	0.950	0.033	0.042	0.001	0.009
<i>No change (β^C)</i>					
Estimate	-0.050 (0.048)	0.005 (0.036)	0.018 (0.034)	0.018 (0.029)	0.039 (0.033)
95% CI	[-0.143, 0.044]	[-0.066, 0.076]	[-0.049, 0.086]	[-0.039, 0.076]	[-0.025, 0.103]
p-value	0.299	0.888	0.593	0.532	0.234
<i>Difference (β^{DIDC})</i>					
Estimate	0.046 (0.074)	0.086 (0.056)	0.044 (0.046)	0.055 (0.037)	0.021 (0.040)
95% CI	[-0.100, 0.192]	[-0.024, 0.195]	[-0.047, 0.134]	[-0.018, 0.128]	[-0.057, 0.099]
p-value	0.536	0.125	0.343	0.141	0.600
N observations (awarded)	3,402	7,434	23,562	47,250	94,374
N observations (no change)	11,718	25,606	81,158	162,750	325,066
N unique reviewers	4,910	10,364	30,676	55,241	66,260
Bandwidth	14.0 days	30.0 days	93.7 days	187.5 days	374.9 days

Notes: Estimates are computed by differencing two local linear regression discontinuity estimators $\beta^A - \beta^C = \beta^{DIDC}$, where β^A comes from awarded restaurants and β^C comes from same region, no change restaurants (no status change but same ceremony date). Both single sample fits use `rdrobust` triangular kernel with local polynomial order $p = 1$. The two fits share the same bandwidth, where h^* is the MSE optimal bandwidth from the awarded sample. Inference on DIDC parameter β^{DIDC} follows Picchetti et al. (2024).

powered than the larger windows with the two week point estimate of -0.618 significant at the 0.05-level and the one month estimate of -0.307 significant at the 0.1-level.

The interpretation of the negative ratings discontinuities we advanced in our main results become more apparent when comparing estimates for heavy reviewers and “casual” reviewers (all other reviewers ex-heavy). The immediate rating response is much more negative and significant at a decline of -0.618 stars relative to the pooled and insignificant -0.290 stars, whose magnitude is also likely driven by that of the heavy reviewers. We hypothesize the expectation difference mechanism discussed briefly in section 5.1 to be exacerbated among heavy reviewers, whose priors are not only informed by the MICHELIN star itself but also by their prior MICHELIN dining experiences. Like for other reviewers and in the pooled sample, the degree of negativity attenuates toward zero as bandwidth choice grows.

The narrative continues through table 7. Despite more experience and individual review tendencies among heavy MICHELIN reviewers, positive and significant estimates from casual reviewers in columns (B) through (E) are larger in magnitude than those from heavy reviewers and may reveal an excitement among casual reviewers to share their new MICHELIN dining experiences online aligned with the hypothesis we present in our conceptual framework. Unlike ratings, casual reviewers appear to be those driving the mag-

Table 6: Effect of Michelin Star-Status Changes on Star Rating — Review-Level Panel, by Reviewer Type

	(A) ± 14 days	(B) ± 30 days	(C) $\frac{1}{2} h^*$	(D) h^*	(E) $2 h^*$
<i>Panel A: Heavy Reviewers</i>					
Estimate	-0.618 (0.284)	-0.307 (0.184)	0.021 (0.100)	-0.013 (0.065)	-0.045 (0.056)
95% CI	[-1.174, -0.061]	[-0.669, 0.054]	[-0.174, 0.217]	[-0.140, 0.113]	[-0.154, 0.065]
p-value	0.030	0.095	0.832	0.836	0.421
N observations	258	543	2,495	4,519	7,746
N unique reviewers	254	511	2,052	3,379	5,026
Bandwidth	14.0 days	30.0 days	129.1 days	258.2 days	516.5 days
<i>Panel B: All Other Reviewers</i>					
Estimate	-0.101 (0.176)	-0.082 (0.147)	0.070 (0.092)	-0.037 (0.062)	-0.013 (0.059)
95% CI	[-0.446, 0.243]	[-0.370, 0.207]	[-0.110, 0.251]	[-0.158, 0.085]	[-0.129, 0.103]
p-value	0.565	0.579	0.445	0.557	0.825
N observations	559	1,206	3,078	6,061	10,850
N unique reviewers	588	1,224	3,017	5,919	10,486
Bandwidth	14.0 days	30.0 days	74.0 days	148.0 days	296.1 days

Notes: Estimates computed using `rdrobust` with triangular kernel and local polynomial order $p = 1$. Standard errors in parentheses are bias-adjusted and clustered at the restaurant level. p -value tests the sharp null of no discontinuity at the cutoff against the two-sided alternative. h^* is the MSE optimal bandwidth. An observation is one individual review whose posting date falls within the bandwidth, and the outcome is the per-review star rating (1–5). Heavy reviewers are diners with five or more reviews across 2015 through 2019 written and posted on a MICHELIN restaurant; all other reviewers are the remaining reviewers in the sample.

nitude of the volume response.

6 Conclusion

In this paper, we explore the effects of MICHELIN award receipt on diner behavior expressed through ratings and review volume. Across all reviewers, receipt of the award causes a negative shock, albeit indistinguishable from zero, in average ratings and a positive, significant shock in review volume. The negative rating response is concentrated among heavy reviewers, those with five or more MICHELIN reviews across our time window, among whom award receipt causes a drop in around -0.618 stars; conversely the positive review volume response is concentrated among casual reviewers, among whom the average number of weekly reviews increases by nearly double that of heavy reviewers, though these effects are still very small (in the pooled sample, the aggregate effect equates to around an additional review each week). Given the power of social media attention, our findings may inform how restaurants of MICHELIN status or caliber adjust their behavior in response to receipt of such a prestigious award to sidestep the negative ratings shock and capture more of the positive review volume shock.

There are several limitations to our exploration. For starters, our data only spans three MICHELIN guides (New York, Chicago, and California) over 5 or so years (2015–2019) with limited information on observable covariates beyond location, prior award status, and

Table 7: Effect of Michelin Star-Status Changes on Review Volume — Daily Panel, by Reviewer Type

	(A) ± 14 days	(B) ± 30 days	(C) $\frac{1}{2} h^*$	(D) h^*	(E) $2 h^*$
<i>Panel A: Heavy Reviewers</i>					
Estimate	-0.002 (0.021)	0.011 (0.017)	0.018 (0.010)	0.017 (0.006)	0.019 (0.006)
95% CI	[-0.044, 0.041]	[-0.023, 0.045]	[-0.001, 0.037]	[0.004, 0.029]	[0.006, 0.031]
p-value	0.944	0.509	0.062	0.009	0.003
N observations	3,402	7,434	28,098	56,070	112,266
N unique reviewers	254	511	1,769	3,038	4,659
Bandwidth	14.0 days	30.0 days	111.5 days	222.9 days	445.9 days
<i>Panel B: All Other Reviewers</i>					
Estimate	-0.006 (0.025)	0.043 (0.018)	0.024 (0.013)	0.027 (0.010)	0.022 (0.009)
95% CI	[-0.056, 0.043]	[0.007, 0.079]	[-0.001, 0.050]	[0.008, 0.047]	[0.003, 0.040]
p-value	0.797	0.019	0.064	0.005	0.022
N observations	3,402	7,434	23,058	45,990	92,106
N unique reviewers	588	1,224	3,722	6,999	12,080
Bandwidth	14.0 days	30.0 days	91.4 days	182.7 days	365.5 days

Notes: Estimates computed using `rdrobust` with triangular kernel and local polynomial order $p = 1$. Standard errors in parentheses are bias-adjusted and clustered at the restaurant level. p -value tests the sharp null of no discontinuity at the cutoff against the two-sided alternative. h^* is the MSE optimal bandwidth. An observation is a (restaurant $\times$ event $\times$ day) cell in the balanced daily panel within the running-variable bandwidth, and the outcome is number of reviews. Heavy reviewers are diners with five or more reviews across 2015 through 2019 written and posted on a MICHELIN restaurant; all other reviewers are the remaining reviewers in the sample.

cuisine type. A sample spanning more geographies over a longer period of time with more controls presents two advantages. First, it enables robustness (e.g., improved precision per Calonico, Cattaneo, Farrell, and Titiunik) and placebo testing (e.g., covariate smoothness tests per Cattaneo et al.) not possible in our setting but empowering in any complete RD application. Second, it facilitates additional heterogeneity analysis (e.g., by star change from 0 to 1 star or 3 to 2 stars) which may reveal more about specific mechanisms through which prestige affects behavior, and for whom.⁹

Finally, our analysis may benefit from alternative or additional specifications. For instance, given the scale of our review data, possible logarithm or other transformations of the review levels may provide more interpretable coefficients beyond the strange level-level estimates we provide in section 5.1.¹⁰ Additionally, we may consider supplementing our existing or further heterogeneity analyses with text sentiment analysis, made accessible via large language models, through which we can more carefully evaluate the degree of rating positivity or negativity by its accompanying review sentiment.

⁹Our attempt at running such a heterogeneity analysis was unfruitful given the breadth of our incomplete panel and review data. All specifications were underpowered, yielding uninformative, insignificant results not worthy of additional comment.

¹⁰Since some restaurants have 0 reviews posted prior to their receipt of their MICHELIN award, a direct log transform would not be appropriate. A peer of mine recommended the one offset log transform or a Poisson regression, but advised caution in interpreting coefficients per Chen and Roth (2024).

A Proof of Proposition 1

Proposition 1. Suppose ex-MICHELIN prestige $\tilde{\psi}_{jt} = \rho \log N_{j,t-1} + \eta R_{j,t-1}$, restaurant quality q_{jt} , and price p_{jt} are continuous at time $t = T_j$. Assume review volume satisfies

$$N_{jt} = g(\psi_{jt}, q_{jt}, p_{jt}) + \nu_{jt} \text{ with } \frac{\partial g}{\partial \psi} > 0,$$

with $\mathbb{E}[\nu_{jt}]$ continuous at T_j . Assume observed ratings satisfy

$$R_{ijt} = q_{jt} + \theta_i \psi_{jt} + \xi_{ijt} \text{ with } \mathbb{E}[\theta_i \mid i \text{ reviews restaurant } j] > 0,$$

with $\mathbb{E}[\xi_{ijt} \mid i \text{ reviews restaurant } j]$ continuous at T_j . Then MICHELIN recognition generates positive discontinuities in expected review volume and expected ratings at the award date. Formally,

$$\begin{aligned} \lim_{t \searrow T_j} \mathbb{E}[N_{jt}] - \lim_{t \nearrow T_j} \mathbb{E}[N_{jt}] &> 0 \\ \lim_{t \searrow T_j} \mathbb{E}[R_{ijt}] - \lim_{t \nearrow T_j} \mathbb{E}[R_{ijt}] &> 0. \end{aligned}$$

Proof. Fix a restaurant j , which is announced as a new MICHELIN award recipient at time $t = T_j$. Define treatment indicator $M_{jt} = \mathbf{1}\{t \geq T_j\}$. Then the only mechanically discontinuous component of prestige at T_j is the term δM_{jt} . Since $\tilde{\psi}_{jt}$ is continuous at T_j , we have that

$$\lim_{t \nearrow T_j} \tilde{\psi}_{jt} = \lim_{t \nearrow T_j} (\tilde{\psi}_{tj} + \delta M_{jt}) = \tilde{\psi}_{jT_j},$$

which follows because $M_{jt} = 0$ for $t < T_j$. Similarly, observe that because $M_{jt} = 1$ for $t \geq T_j$, we have:

$$\lim_{t \searrow T_j} \tilde{\psi}_{jt} = \lim_{t \searrow T_j} (\tilde{\psi}_{tj} + \delta M_{jt}) = \tilde{\psi}_{jT_j} + \delta.$$

Therefore the jump in perceived prestige at the award date is precisely

$$\lim_{t \searrow T_j} \psi_{jt} - \lim_{t \nearrow T_j} \psi_{jt} = \delta > 0, \tag{*}$$

and MICHELIN recognition causes a discontinuous increase in perceived prestige of size δ . Now consider the result for expected review volume. By assumption,

$$N_{jt} = g(\psi_{jt}, q_{jt}, p_{jt}) + \nu_{jt}.$$

Assume $\mathbb{E}[\nu_{jt}]$ is continuous at $t = T_j$ with no ex-MICHELIN discontinuous shock in the review volume error term exactly at $t = T_j$. Because the error term contributes no discontinuous jump in expectation, we have:

$$\lim_{t \searrow T_j} \mathbb{E}[N_{jt}] - \lim_{t \nearrow T_j} \mathbb{E}[N_{jt}] = \lim_{t \searrow T_j} \mathbb{E}[g(\psi_{jt}, q_{jt}, p_{jt})] - \lim_{t \nearrow T_j} \mathbb{E}[g(\psi_{jt}, q_{jt}, p_{jt})].$$

By continuity of q_{jt} and p_{jt} at T_j , we have both that:

$$\begin{aligned}\lim_{t \searrow T_j} q_{jt} &= \lim_{t \nearrow T_j} q_{jt} = q_{jT_j} \\ \lim_{t \searrow T_j} p_{jt} &= \lim_{t \nearrow T_j} p_{jt} = p_{jT_j}.\end{aligned}$$

Therefore the only discontinuous argument of $g(\cdot)$ is ψ_{jt} . From $(\star)$, the left hand limit is $g(\tilde{\psi}_{jT_j}, q_{jT_j}, p_{jT_j})$ and the right hand limit is $g(\tilde{\psi}_{jT_j} + \delta, q_{jT_j}, p_{jT_j})$. Together,

$$\lim_{t \searrow T_j} \mathbb{E}[N_{jt}] - \lim_{t \nearrow T_j} \mathbb{E}[N_{jt}] = g(\tilde{\psi}_{jT_j}, q_{jT_j}, p_{jT_j}) - g(\tilde{\psi}_{jT_j} + \delta, q_{jT_j}, p_{jT_j}) > 0,$$

where the inequality follows from the fact $g(\cdot)$ is increasing in prestige. Finally we consider ratings, for which the observed review equation is

$$R_{ijt} = q_{jt} + \theta_i \psi_{jt} + \xi_{ijt}.$$

Taking expectations conditional on reviewer i reviewing restaurant j :

$$\mathbb{E}[R_{ijt} | i \text{ reviews } j] = q_{jt} + \psi_{jt} \mathbb{E}[\theta_i | i \text{ reviews } j] + \mathbb{E}[\xi_{ijt} | i \text{ reviews } j].$$

Assuming $\mathbb{E}[\xi_{ijt} | i \text{ reviews } j]$ is continuous at T_j similarly to $\mathbb{E}[\nu_{jt}]$. q_{jt} is also continuous at T_j , so the only discontinuous component of expected ratings is the prestige term. It follows that:

$$\lim_{t \searrow T_j} \mathbb{E}[R_{ijt}] - \lim_{t \nearrow T_j} \mathbb{E}[R_{ijt}] = \mathbb{E}[\theta_i | i \text{ reviews } j] \left[\lim_{t \searrow T_j} \psi_{jt} - \lim_{t \nearrow T_j} \psi_{jt} \right].$$

We have already shown in $(\star)$ the bracketed term is $\delta > 0$. Since $\mathbb{E}[\theta_i | i \text{ reviews } j] \geq 0$ by assumption, their product is also non-negative. This proves the ratings result. Note if $\frac{\partial g}{\partial \psi} > 0$ locally around $\psi = \tilde{\psi}_{jT_j}$, then $g(\tilde{\psi}_{jT_j} + \delta, q_{jT_j}, p_{jT_j}) > g(\tilde{\psi}_{jT_j}, q_{jT_j}, p_{jT_j})$, so the review volume discontinuity is strictly positive. The result holds similarly if $\mathbb{E}[\theta_i | i \text{ reviews } j] > 0$.

Thus MICHELIN recognition generates weakly positive discontinuities in expected review volume and expected ratings at the award date, and strictly positive discontinuities under strict monotonicity and strictly positive average reviewer prestige perception. $\square$

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